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Antiaromatic Covalent Organic Frameworks Based on Dibenzopentalenes

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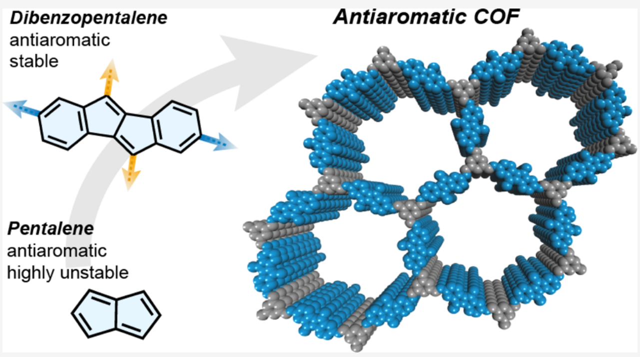

ABSTRACT: Despite their inherent instability, 4*n* π systems have recently received significant attention due to their unique optical and electronic properties. In dibenzopentalene (DBP), benzannellation stabilizes the highly antiaromatic pentalene core, without compromising its amphoteric redox behavior or small HOMO−LUMO energy gap. However, incorporating such molecules in organic devices as discrete small molecules or amorphous polymers can limit the performance (e.g., due to solubility in the battery electrolyte solution or low internal surface area). Covalent organic frameworks (COFs), on the contrary, are highly ordered, porous, and crystalline materials that can provide a platform to align molecules with specific properties in a well-defined, ordered environment. We synthesized the first antiaromatic framework materials and obtained a series of three highly crystalline and porous COFs based on DBP. Potential applications of such antiaromatic bulk materials were explored: COF films show a conductivity of 4×10^{-8} S cm^{-1} upon doping and exhibit photoconductivity upon irradiation with visible light. Application as positive electrode materials in Li-organic batteries demonstrates a significant enhancement of performance when the antiaromaticity of the DBP unit in the COF is exploited in its redox activity with a discharge capacity of 26 mA h g^{-1} at a potential of 3.9 V vs. Li/Li$^+$. This work showcases antiaromaticity as a new design principle for functional framework materials.

INTRODUCTION

In the field of organic materials, polycyclic aromatic molecules have received much attention owing to their electronic properties. According to Hückel's rule, aromatic molecules are planar, monocyclic systems with cyclic conjugated [4*n* + 2] π-electrons resulting in substantial stabilization. On the contrary, planar cyclic systems with 4*n* conjugated π-electrons experience destabilization and are termed antiaromatic.[1] Antiaromaticity comes with intriguing properties, such as a low HOMO−LUMO energy gap,[2] a low-lying triplet state energy,[3−5] and redox activity.[6−10] However, utilizing these properties is hampered by the inherent instability of such systems. Annellation of aromatic rings to (polycyclic) antiaromatic molecules[11−14] can stabilize the reactive cores thermodynamically by adding local aromaticity to the system while retaining the peculiar properties ascribed to antiaromaticity.[9,15−19] For example, benzannellation to the highly antiaromatic and unstable pentalene[20] leads to dibenzo[*a*,*e*]-pentalene (DBP),[21−26] which is benchtop stable despite its 4*n* π-electron count (Figure 1a). DBP and related pentalene-based compounds show a low and tunable HOMO−LUMO energy gap[26] and a higher stability than acenes of similar size.[27] Thanks to these properties, interest in such polycyclic 4*n* π molecules for organic optoelectronics, such as organic field-effect transistors (OFETs)[7,28−30] or organic photovoltaics,[31−33] has surged recently. Another feature directly related to their antiaromatic nature is a reversible ambipolar redox behavior, unusual for pure hydrocarbons. When 4*n* π-systems are oxidized or reduced, the molecules experience aromatic stabilization in both the doubly reduced and doubly oxidized states, resulting in an aromatic [4*n* + 2] π-electron count.[34,35] Combined with their comparably high HOMO and low LUMO energy, this renders 4*n* π compounds potentially interesting candidates for functional materials,[22] such as organic field-effect transistors or batteries.

The field of organic electroactive materials is still underexplored, and mostly well-established building blocks have been

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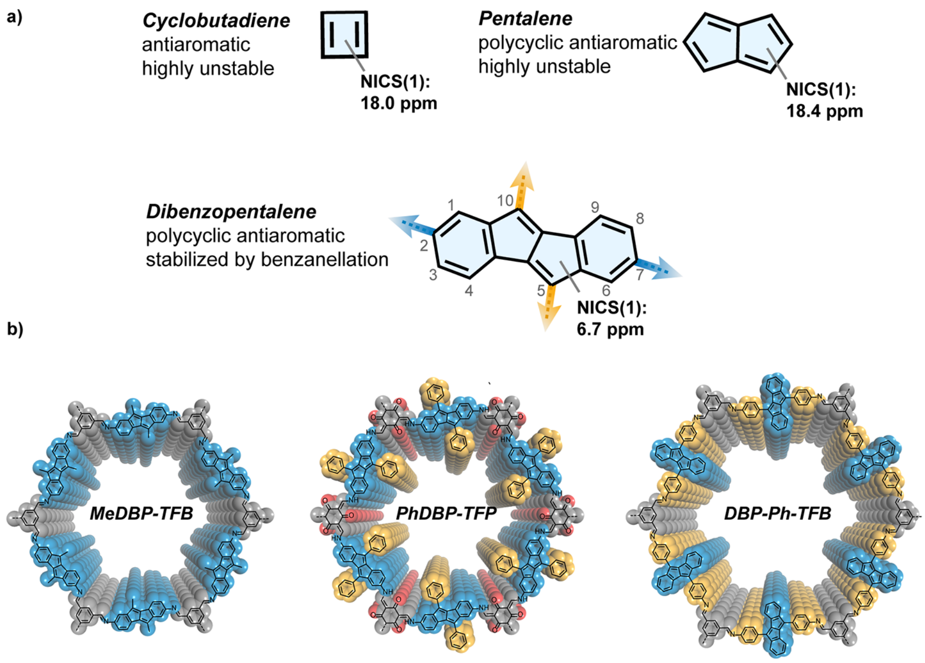


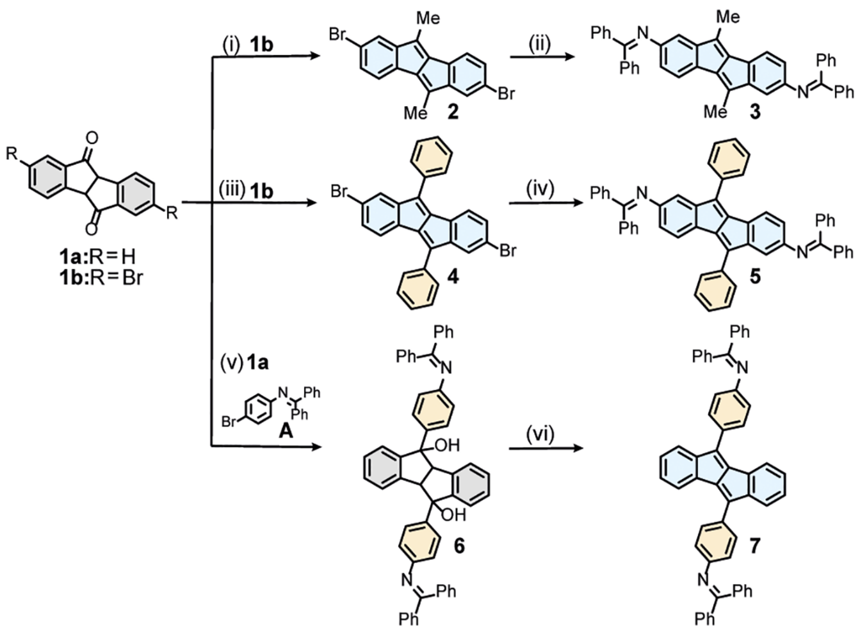
Figure 1. (a) Structure of antiaromatic cyclobutadiene, pentalene, and dibenzo[*a*,*e*]pentalene, with $NICS_{iso}(1)$ values [DFT, mPW1PW91/6-311+G(2d,p)] indicating an attenuated antiaromaticity in dibenzopentalene compared to pentalene; (b) structures of antiaromatic COFs presented in this work.

employed so far. Conceptually, new redox systems are still to be explored to expand the field and pave the way for future applications. In this regard, antiaromatic building blocks are interesting, with their redox-activity emerging from the pure hydrocarbon scaffold. With their ambipolar redox behavior, antiaromatic building blocks can potentially be employed as positive and negative electrode materials, also making all-organic batteries feasible. Therefore, exploring antiaromatic building blocks as electroactive units in energy storage materials is of high interest. However, to this day, examples of such antiaromatic energy storage materials are rare. One example was reported by Shinokubo and co-workers, employing antiaromatic norcorrole as an electrode-active material in a battery.[8] Examples are scarce because of the limiting structural material design as discrete small molecules or amorphous polymers in electrode-active materials posing performance problems due to solubility in the electrolyte solution, instability in the charged state, or a low internal surface area for efficient ion diffusion.[36−38]

Some of these issues are addressed by porous framework materials, such as covalent organic frameworks (COFs)[39,40] and porous organic polymers (POPs),[41−44] emerging classes of advanced materials. These highly porous materials are constructed from functional monomers and can serve as a platform to translate the properties of molecules to bulk insoluble materials. In contrast to amorphous POPs, COFs are crystalline materials. Thanks to their high degree of order, high surface area, accessible pores, and efficient electronic delocalization, COFs are ideal candidates for applications in energy storage[45−47] as well as gas storage and separation,[48,49] (photo)catalysis,[50−53] and optoelectronics.[54−62] The architecture of 2D COFs with their uniform channels, high stability, and insolubility is especially ideal for applications as electrode-active material in batteries.[45,47,63−69] So far, mostly well-established organic redox units have been employed as active centers in COF-based electrode-active materials.[45] Their redox activity is based on the presence of discrete redox centers, such as quinone groups, containing heteroatoms. On the other hand, antiaromatic molecules can consist solely of carbon and hydrogen atoms but still show defined redox activity based on the aromatic stabilization of charged states. They have not yet been incorporated into COFs although such antiaromatic frameworks are an interesting target both as materials and from a fundamental perspective.[70] The ambipolar redox properties, together with the possibility of charge delocalization through close $\pi-\pi$ stacking interactions in antiaromatic units,[71−73] hold potential for specific material properties of interest for energy storage materials. The highly porous, crystalline COF can provide channels for ion diffusion and an insoluble, robust framework to incorporate antiaromatic building blocks as conceptually novel redox-active units based on the interplay of aromaticity and antiaromaticity.

In this work, we designed a series of imine-linked 2D COFs incorporating DBPs as antiaromatic subunits. We chose DBP as a COF building block for its balance between stability and interesting properties based on its antiaromatic character. Nucleus-independent chemical shift (NICS)[74] values show a decreased antiaromaticity compared to the parent pentalene (Figure 1a).[75−77] Polycyclic $4n$ π compounds with more pronounced antiaromatic character would hamper COF formation due to their limited thermal stability under the harsh reaction conditions used in reticular chemistry (typically 120 °C, 3 d). Despite their moderate antiaromaticity, DBP and its derivatives show remarkable redox properties and low, tunable band gaps, typical for $4n$ π systems.[16,26] We obtained three highly crystalline COFs (Figure 1b) and two amorphous POPs, all highly porous and stable materials based on DBP as a $4n$ π building block. Antiaromatic COF films were used to

Scheme 1. Synthesis of DBP-Based COF Building Blocks[a]



[a]Reagents and conditions: (i) (1) $CeCl_3$, CH_3MgBr, **1b**, 0→25 °C, 4 h; (2) *p*TsOH, toluene, 110 °C, 4 h, 67%; (ii) $Pd(dba)_2$, BINAP (*rac*), NaO*t*Bu, benzophenone imine, 100 °C, 2 h, 90%; (iii) (1) $CeCl_3$, PhMgBr, **1b**, 0 → 25 °C, 20 min; (2) *p*TsOH, toluene, 110 °C, 4 h, 54%; (iv) $Pd(dba)_2$, BINAP (*rac*), NaO*t*Bu, benzophenone imine, 100 °C, 1 h, 79%; (v) **A**, *n*BuLi, then **1a**, THF, −78 → 25 °C, 3 h, 68%; (vi) Burgess reagent, toluene, 110 °C, 1 h, 89%.

explore the low optical band gap of the material as a photoresponsive semiconductor with a conductivity of 4×10^{-8} S cm^{-1} after doping. We assembled battery cells with one of the antiaromatic DBP-COFs and one POP as electrode-active materials and investigated their charge storage properties in Li-based organic half-cell batteries. When employed as positive electrode materials, oxidation occurs at high potentials of ca. 3.9 V vs. Li/Li^+ with specific capacities of up to 30 mA h g^{-1}. When utilizing the antiaromatic COF as negative electrode materials, Li-ions are inserted at potentials around 0.5 V vs. Li/Li^+ with specific capacities of above 200 mA h g^{-1}. We show that the intrinsic ambipolar redox properties of the antiaromatic DBP units are transferred to the bulk materials for their application electrode materials as either negative or positive electrodes in rechargeable batteries.

RESULTS AND DISCUSSION

Design and Synthesis of COF Building Blocks. We identified two possible linkage geometries—the 2,7-substitution and 5,10-substitution—through which DBP can be incorporated into a COF as a (pseudo-)linear linker (Figure 1a).[78,79] To access imine-linked COFs, we installed benzophenone-protected amines at those positions. The benzophenone-imine groups serve a dual function: they are readily accessible via Buchwald−Hartwig imination reaction[80] from the corresponding bromides and are less prone to oxidation than the respective free diamines of DBPs that proved unstable in all our initial attempts. Second, benzophenone imines can be used directly in COF synthesis, improving the porosity of COFs compared to employing free amines.[81] In a formal transimination reaction, an imine- or ketoenamine-connected COF is formed, and benzophenone is cleaved off.

For a 2,7-bisfunctionalized DBP, the Grignard addition of CH_3MgBr to dibromodiketone **1b**[75] promoted by $CeCl_3$ led to a diol that was subsequently dehydrated under acidic conditions, affording the dimethyl-substituted DBP **2** in a yield of 67% (Scheme 1).[78] Buchwald−Hartwig imination of dibromide **2** using benzophenone imine as a coupling partner furnished dimethyl-bis(benzophenone imine) **3** in 90% yield. The same strategy was applied to achieve a different substituent: the corresponding diphenyl-substituted COF linker **5** was synthesized analogously via dibromide **4**. To achieve a 5,10-linked DBP monomer, a different approach was necessary: for the synthesis of monomer **7**, a Li-organyl bearing the benzophenone imine unit was prepared from bromide **A**. The Li-organyl was reacted with diketone **1a**[75] to afford diol **6** in 68% yield. Dehydration of diol **6** without hydrolyzing the imine functionalities was achieved using Burgess's reagent, yielding COF-linker **7** in 89% yield.[82] An attempted synthesis of **7** via Buchwald−Hartwig coupling of a 5,10-(4-bromophenyl)-DBP **S1** was not successful as the reaction led to an inseparable mixture (debromination products among others). 5,10-Linked bis(benzophenone imine) **7** is hardly soluble in common organic solvents, making even ^{1}H NMR spectroscopy challenging. This is unusual as its 5,10-bromophenyl-substituted analogue **S1** is sufficiently soluble, and typically benzophenone imine substitution enhances solubility.[81]

The optical, electrochemical, and structural properties of **2**−**7** were consistent with previous reports of DBPs.[16,26,28,75,76] The lowest energy transition in DBPs is symmetry forbidden (HOMO → LUMO) and hardly visible in the UV/vis absorption spectra.[15] The spectra of the DBP imines **3**, **5**, and **7** show a bathochromic shift compared to the respective DBP bromides **2**, **4**, and **S5**. Cyclic voltammetry shows the amphoteric redox behavior of the DBPs (Supporting Information, Section S16). DBPs **2** and **3** with methyl groups in the 5,10-positions show irreversible reduction and oxidation waves, while 5,10-phenyl substitution (in **4** and **5**) leads to an

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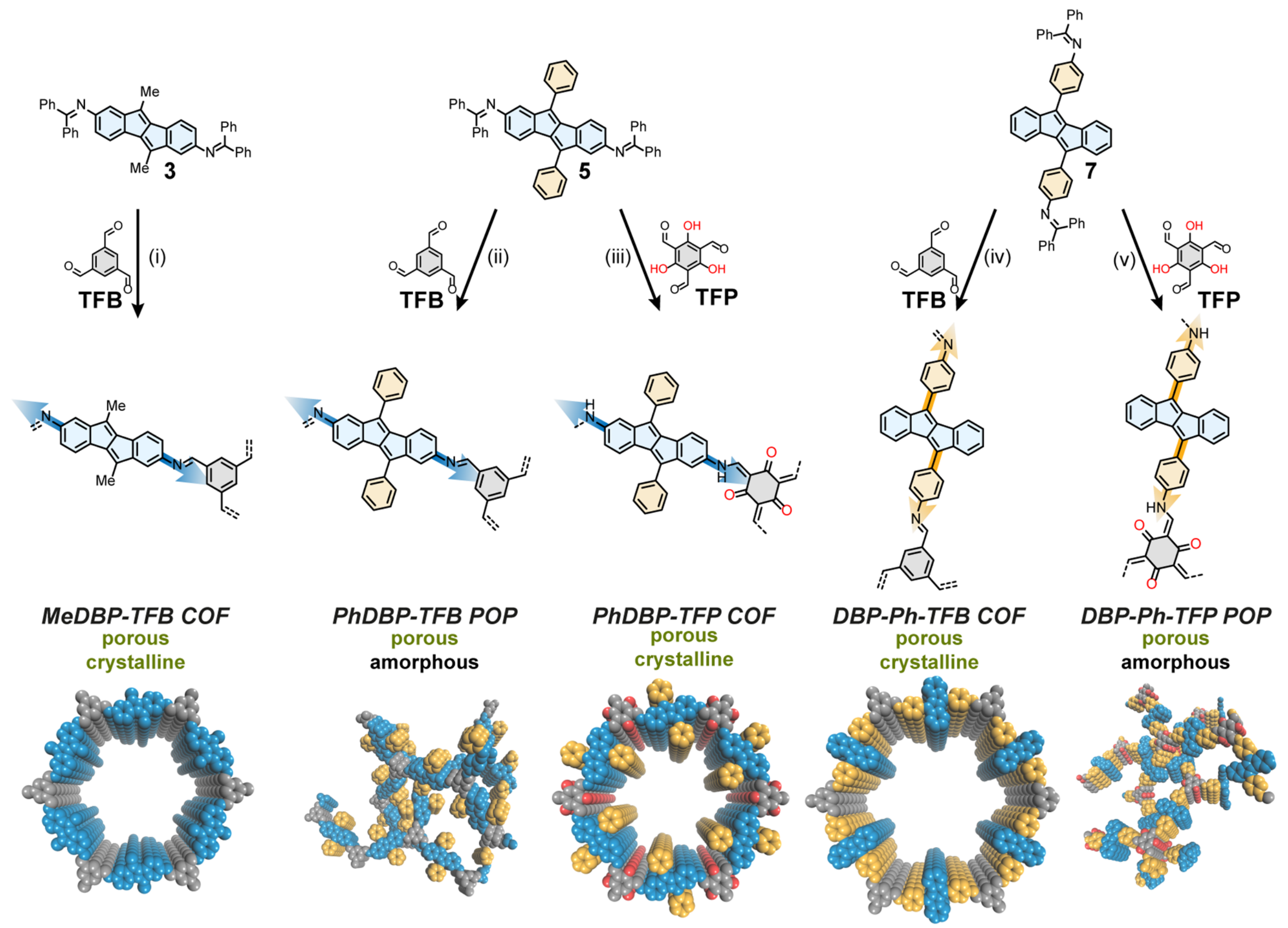

Figure 2. Top: construction of DBP-based COFs and POPs from linearly functionalized DBPs **3**, **5**, and **7** with trigonal nodes **TFB** and **TFP**. Reagents and conditions (i−v): acid, mesitylene/1,4-dioxane 1:1, 120 °C, 3 d, with the following specifications: (i) aq. TFA (6 M, 18 equiv), 88%; (ii) aq. TFA (6 M, 18 equiv), 92%; (iii) aq. TFA (8 M, 28 equiv), 77%; (iv) trifluoromethanesulfonic acid (conc., 6 equiv), 71%; (v) aq. TFA (6 M, 18 equiv), 76%. Bottom: Cartoon representations of crystalline COFs and amorphous POPs.

improved (pseudo)-reversible reduction and oxidation. The antiaromatic character of the target compounds is reflected in their nuclear magnetic resonance (NMR) spectra: in line with reported DBPs, the ^{1}H NMR signals of the benzenic protons at the DBP appear upfield shifted (6.15−7.14 ppm) relative to ^{1}H NMR signals of fully aromatic molecules (naphthalene: 7.47−7.84 ppm). $NICS_{iso}(1)$ calculations [level of theory: DFT:mPW1PW91/6-311G+(2d,p)] reveal paratropicity in the 5-membered rings and diatropicity in the 6-membered rings, analogous to reported DBP derivatives[75] [e.g., for **5**: $NICS_{iso}(1)_{5\text{-ring}} = 6.26$ ppm, $NICS_{iso}(1)_{6\text{-ring}} = -4.61$ ppm; see the Supporting Information, Section S16]. Single crystals suitable for X-ray crystallography were grown for DBPs **2**, **4**, and **7**. They show bond length alternations typical for compounds with antiaromatic character and a herringbone packing typical for DBPs (Supporting Information, Section S17).

COF Synthesis and Characterization. For COF construction, we reacted the three DBP-based linear linkers **3**, **5**, and **7** with commercially available triformylbenzene (**TFB**), forming an imine bond, or with triformylphloroglucinol (**TFP**), forming a keto-enamine by keto−enol-tautomerization after imine bond formation. After solvothermal synthesis, we obtained three crystalline porous COFs and two amorphous porous polymers (POPs, Figure 2).

MeDBP-TFB COF (Figure 3a) was synthesized under solvothermal conditions (mesitylene/1,4-dioxane 1:1, 18 equiv trifluoroacetic acid (TFA), 120 °C, 3 d). Powder X-ray diffraction (PXRD) showed narrow reflexes, revealing structural features and confirming the formation of a highly crystalline framework (Figure 3b). Comparison with a simulated diffraction pattern from structural model using the Materials Studio program package[83] showed agreement with an eclipsed AA stacking pattern. Pawley refinement in the hexagonal $P6/m$ space group provided a good fit in agreement with the experimental data (refined lattice parameters $a = b = 32.5$ Å and $c = 3.8$ Å). The most intense reflex was observed at $2\theta = 3.3°$ corresponding to the (100) Bragg reflection as well as a reflex at 6.5° corresponding to the (200) plane. The planes (110) and (120) are represented in more weakly resolved reflexes at 5.7 and 8.7°, respectively. A broad reflex was observed at 25.8° corresponding to the (001) plane and confirming the $\pi-\pi$ stacked 2D structure at a layer distance of 3.8 Å. Nitrogen (N_2) adsorption−desorption measurements at 77 K confirmed permanent porosity of **MeDBP-TFB** COF (Figure 3c). A steep initial increase in $p/p_0 = 0-0.02$ suggests a type I isotherm, confirming microporosity of the framework.[84]

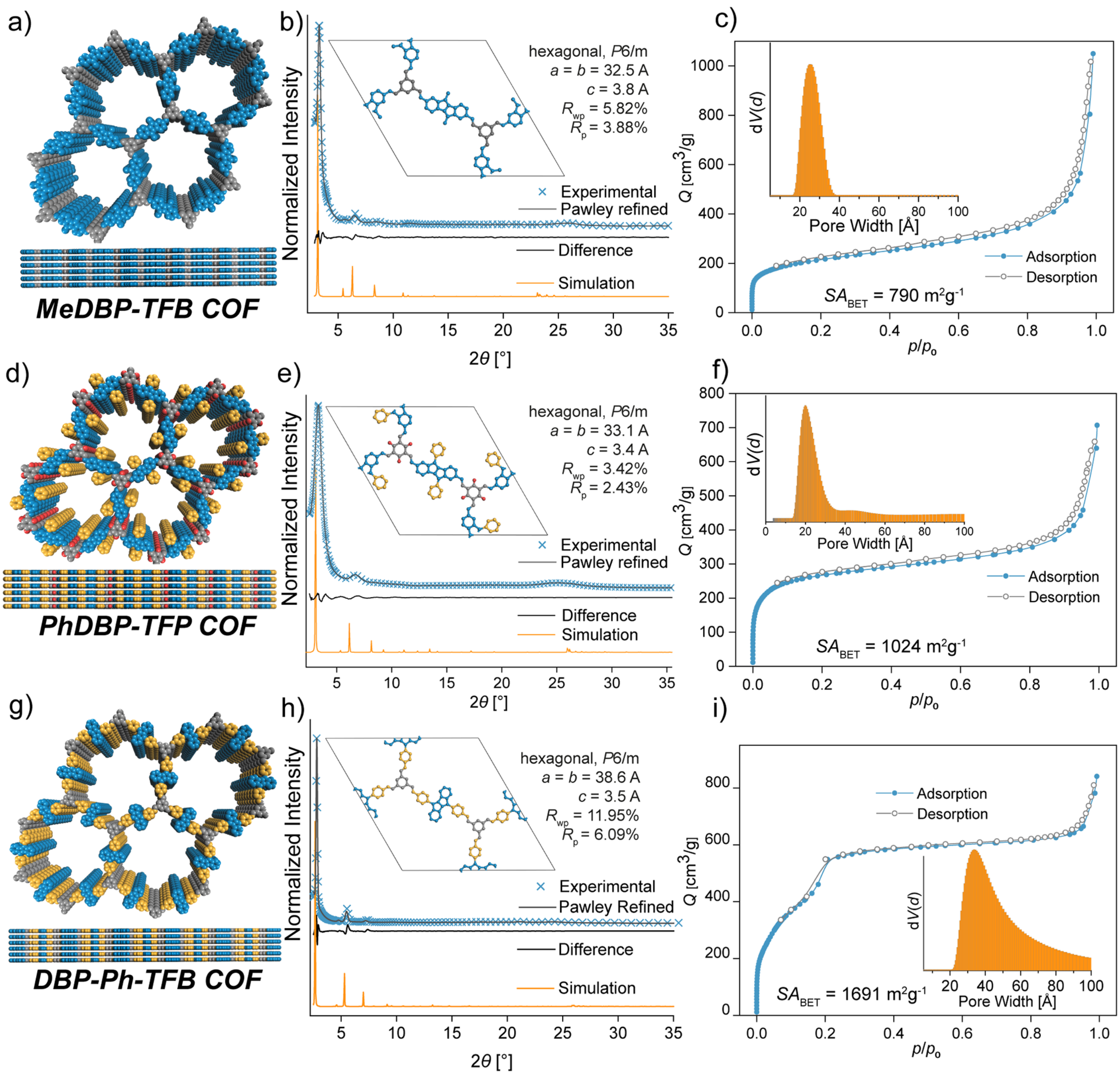

Figure 3. Characterization of DBP COFs. (a,d,g) Structural model of **MeDBP-TFB** COF, **PhDBP-TFP** COF, and **DBP-Ph-TFB** COF; (b,e,h) PXRD data of **MeDBP-TFB** COF, **PhDBP-TFP** COF, and **DBP-Ph-TFB** COF: experimental PXRD pattern (blue crosses), Pawley refinement (gray line) using an optimized structural model (inset), difference between Pawley refinement and experimental data (black line), simulated PXRD pattern (orange); (c,f,i) N_2 sorption isotherms of **MeDBP-TFB** COF, **PhDBP-TFP** COF, and **DBP-Ph-TFB** COF: N_2 adsorption isotherm (blue), N_2-desorption isotherm (gray), inset: pore size distribution simulated by NL-DFT based on sorption data.

The surface area was calculated using the Brunauer−Emmet−Teller (BET) method, showing a high porosity of 790 $m^2\ g^{-1}$. The pore size distribution within the material was calculated from the adsorption isotherm using nonlinear density functional theory (NL-DFT), showing a narrow distribution of pore sizes centered at around 25 Å (Figure 3c, inset). Examination of the morphology using scanning electron microscopy (SEM) revealed highly uniform aggregates of spherical particles of with a diameter of ∼1 μm (Figure 4a,b). Fourier-transform infrared (FTIR) spectroscopy and cross-polarization magic-angle spinning nuclear magnetic resonance (CP-MAS NMR) were used to examine the chemical composition of **MeDBP-TFB** COF (Supporting Information, Figures S26 and S31). Complete formation of the imine bond was confirmed by the absence of the C═O stretch of **TFB** at 1695 cm^{-1}, while the C═N imine stretch was detected at 1595 cm^{-1}. The CP-MAS NMR spectrum showed a peak at 9.5 ppm corresponding to the CH_3-group, along with overlapping peaks from 120 to 153 ppm corresponding to the aromatic carbons and the imine carbon, as expected in the same area. The absence of an aldehyde signal around 200 ppm again indicated full conversion of the monomers. Thermal and chemical stability were confirmed by TGA and stability tests in various solvents (Supporting Information, Section S12, S13).

Attempts to reticulate phenyl-substituted DBP **5** with **TFB** to a COF (mesitylene/1,4-dioxane 1:1, 18 equiv TFA) resulted

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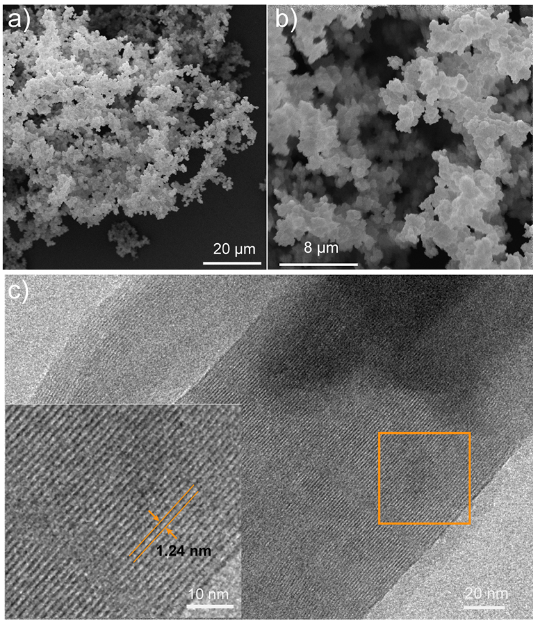

Figure 4. (a,b) SEM image of **MeDBP-TFB** COF bulk powder sample showing uniform particles; see the Supporting Information for analogous images of all other COFs and POPs (Supporting Information, Section S10). (c) Low-dose HR-TEM image of a **PhDBP-TFP** COF crystallite (~500 × 130 nm) featuring lattice fringes with a distance of 1.24 nm corresponding to the (200) reflex in the PXRD; inset: zoom-in.

in a non-crystalline porous polymer **PhDBP-TFB** POP. Even though the material showed high porosity with a BET surface area of 669 m^2 g^{-1}, it proved to be fully amorphous in PXRD analysis (Supporting Information, Figures S3, S4, and S19). In an extensive screening (Table S2), no crystalline COFs but only amorphous porous polymers were produced. Despite its lack of crystallinity, the material showed a narrow pore size distribution calculated from the N_2-adsorption isotherm by NL-DFT, a uniform particle size under SEM, and narrow signals in CP-MAS NMR spectroscopy (Supporting Information, Figures S12, S27, and S42). To extend our screening, we replaced **TFB** by triformylphloroglucinol (**TFP**) as a node. Using **TFP**, the newly formed imine linkage was transformed to a β-ketoenamine by irreversible keto−enol tautomerization, often leading to improved stability compared to COFs containing the **TFB** node.[85] To our delight, the reaction of DBP **5** and **TFP** (mesitylene/1,4-dioxane 1:1, 28 equiv TFA) yielded crystalline **PhDBP-TFP** COF (Figure 3d) as confirmed by PXRD (Figure 3e). The experimental data matched well with the X-ray diffraction pattern calculated from an AA-stacked model. Pawley refinement in the *P*6/*m* space group provided a good fit with the experimental data, yielding refined lattice parameters of $a = b = 33.1$ Å and $c = 3.4$ Å. The reflex corresponding to the (100) plane was observed as the most intense signal at 3.3° as well as the (200) and (001) reflex at 6.7 and 25.3°, respectively. N_2 sorption showed a type I isotherm, and BET analysis revealed a high internal surface area of 1024 m^2 g^{-1} (Figure 3f). Using NL-DFT, a narrow pore size distribution centered around 20 Å was derived from the adsorption isotherm (Figure 3f inset). Low-dose high-resolution transmission electron microscopy (HR-TEM) revealed uniform rod-shaped crystallites and consistent lattice fringes with 1.24 nm spacing, indicative of the crystalline nature of the material (Figure 4c, Section S9). Using Bragg's equation, the d-spacing correlates to a diffraction angle of 2θ = 7.3°, corresponding to the (200) plane, as indicated by the Pawley refinement of the experimental PXRD data. The CP-MAS NMR spectrum showed a peak at 184.0 ppm, corresponding to the ketone carbon, next to an overlapping set of signals from 100 to 150 ppm corresponding to the aromatic and enamine carbons (Supporting Information, Figure S28). FTIR spectroscopy showed the characteristic C═O stretch at 1573 cm^{-1}, confirming the ketoenamine form of the linkage (Supporting Information, Figure S33).

Using DBP as a COF linker through its 5- and 10-position, DBP **7** was reacted with **TFB** and **TFP**, respectively. Here, to our surprise, only the condensation with the **TFB** node yielded crystalline **DBP-Ph-TFB** COF (Figure 3g), while reticulating the same monomer **7** with **TFP** yielded amorphous **DBP-Ph-TFP** POP, in contrast to our findings with DBP linker **5**. The **DBP-Ph-TFB** COF was synthesized in mesitylene/1,4-dioxane 1:1 using 6 equiv of trifluoromethanesulfonic acid. PXRD shows narrow reflexes at 2.8, 4.8, 5.5, and 7.3°, corresponding to the (100), (110), (200), and (120) planes, respectively (Figure 3h). The experimental data agree with the diffraction pattern of an AA-stacked model, and Pawley refinement in the *P*6/*m* space group gave lattice parameters of $a = b = 38.6$ Å and $c = 3.5$ Å. The N_2-sorption isotherms show a clear step around $p/p_0 = 0.2$, indicating a Type IV isotherm and mesoporosity in accordance with the expected pore size, larger than **MeDBP-TFB** COF and **PhDBP-TFP** COF (Figure 3i). In contrast, non-crystalline **DBP-Ph-TFP** POP does not show a step around $p/p_0 = 0.2$, typical for a type IV isotherm, likely indicating an overall lower degree of order (Supporting Information, Figure S9). The materials exhibit a BET surface area of 1691 and 871 m^2 g^{-1} for **DBP-Ph-TFB** COF and **DBP-Ph-TFP** POP, respectively. The pore size distribution calculated by NL-DFT from the adsorption isotherm shows a narrow range around 38 Å (Figure 3i inset).

Optical and Electronic Properties. The optical properties of the antiaromatic COFs and POPs were studied using diffuse reflectance UV−vis spectroscopy. The spectra of the powders revealed broad absorptions with onsets in the visible range beyond 700 nm (Figure 5). **PhDBP-TFP** COF shows the most red-shifted absorption onset. The optical band gaps were determined by Tauc plots and range from 1.3 to 1.7 eV, with **PhDBP-TFP** COF having the smallest band gap. The

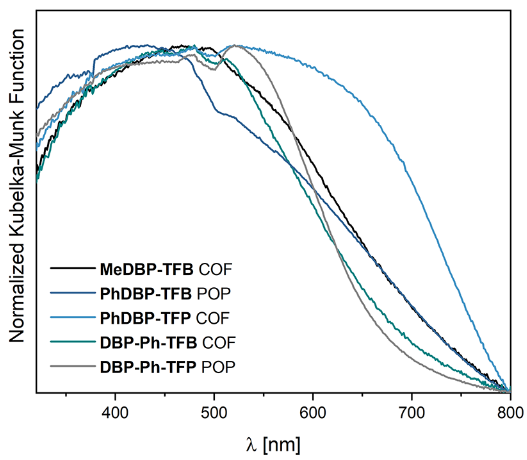

Figure 5. Normalized diffuse reflectance UV−vis spectra of synthesized COF and POP powders.

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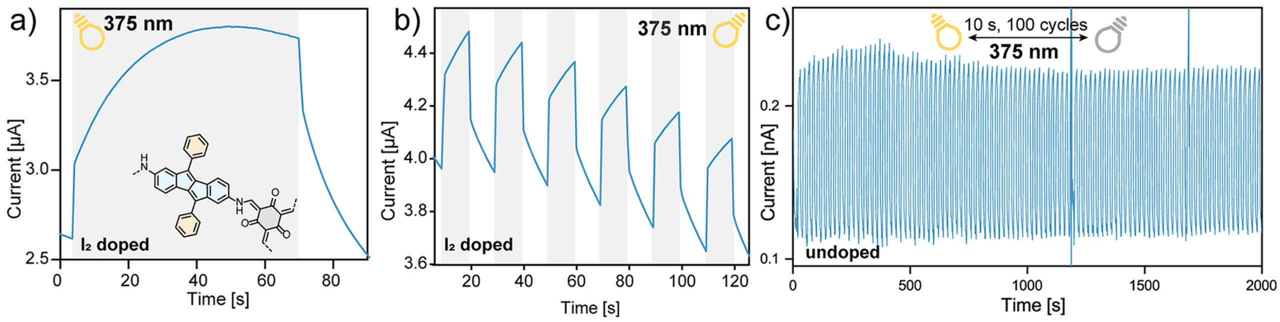

Figure 6. (a) Conductivity of an I_2-doped **PhDBP-TFP** COF film at 20 V bias under irradiation with a 375 nm LED (gray background); (b) conductivity of I_2-doped **PhDBP-TFP** COF film at 20 V bias with 10 s irradiation pulses by a 375 nm LED (gray shaded); (c) conductivity of the **PhDBP-TFP** COF film at 20 V bias with 10 s irradiation pulses by a 375 nm LED over 100 cycles.

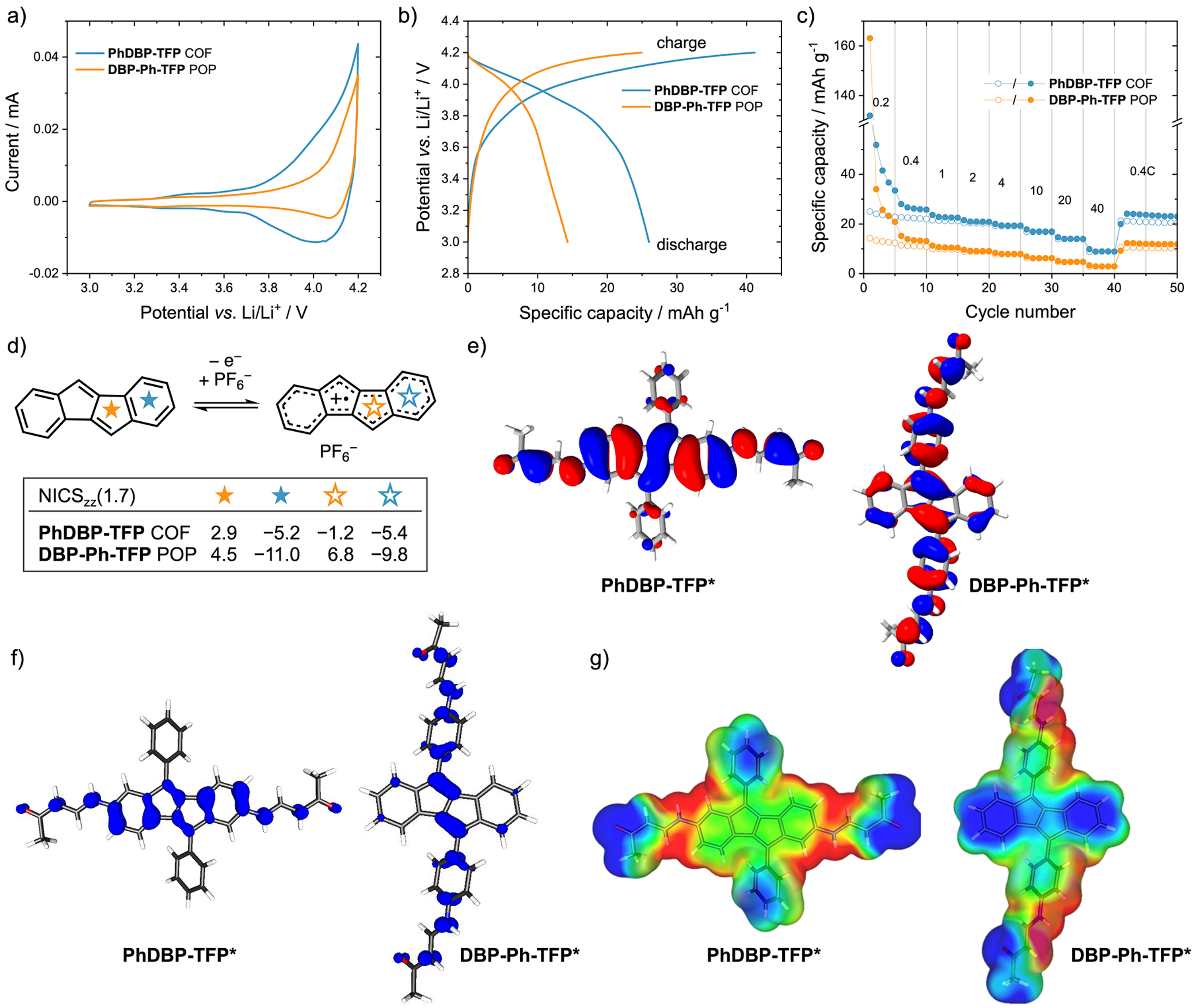

Figure 7. (a−c) Electrochemical performance of **PhDBP-TFP** COF and **DBP-Ph-TFP** POP in Li-organic batteries (COF or POP/acetylene black/PVdF binder (30:50:20 wt %) on Al foil, 1 M $LiPF_6$ in EC/DMC (1:1), counter electrode: Li foil). (a) Cyclic voltammograms (2nd cycle, 0.2 mV s^{-1}); (b) charge/discharge curves during constant current cycling at 100 mA g^{-1}, 2nd cycle; (c) rate performance; (d) redox reaction showing the oxidation of the DBP units to a radical cation during battery measurements and NICS values over the centers of the rings; (e) calculated HOMOs of subunits of **PhDBP-TFP** COF and **DBP-Ph-TFP** POP; (f) calculated spin densities of the radical cation species; (g) calculated electrostatic potential maps of the radical cation species; on blue surfaces, a negative point charge is repelled; on red surfaces, it is attracted. (All calculations were done with PW6B95-D3/def2-QZVP(COSMO)//B3LYP-D3/def2-TZVP(COSMO)).

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J. Am. Chem. Soc. 2023, 145, 2840−2851

trends observed in the diffuse reflectance UV−vis spectra matched with UV−vis spectra of COF dispersions, with **PhDBP-TFP** COF having the most red-shifted absorption maximum.

To investigate the intrinsic conductivity of the material, we fabricated COF thin films. **PhDBP-TFP** COF was chosen for its smallest optical band gap of this series (Figure 5). Dispersions were produced by ultrasonication and subsequent dropcasting onto an interdigitated indium tin oxide (ITO) substrate. Applying a voltage bias of 20 V, the COF showed a low conductivity of 1×10^{-12} S cm^{-1}. p-Type doping by exposure to I_2 vapor (15 h, room temperature) increased the conductivity by 4 orders of magnitude to 4×10^{-8} S cm^{-1} (Figure 6a,b), while n-type doping using tetrakis-(dimethylamino)ethylene (TDAE) was unsuccessful. Upon irradiation with UV or visible light, both doped and undoped films show photoconductivity that can be rationalized by (HOMO−1) → LUMO, HOMO → (LUMO+1) and presumably also from the low-intensity HOMO → LUMO transitions generating charge carriers. While irradiating with 375, 405, 505, or 660 nm, the conductivity approximately doubled (Figure 6a). Even after 100 on/off cycles (10 s on, 10 s off), the material showed no signs of fatigue (Figure 6c). The **PhDBP-TFP** COF films showed an equal on/off ratio in photoresponsive conductivity in the doped and undoped states. Overall, the conductivity is most likely limited not by properties native to DBP-COFs but by the poor film quality. Next to the heterogeneous film morphology, structural disorder at crystal domain interfaces of the nanocrystalline material is likely to hamper efficient charge transport. Similarly, the slow response time/long carrier lifetime is likely a feature of the material quality and not an intrinsic property of the DBP functionalities. Processing as well as growing COFs into homogeneous highly crystalline thin film remains a challenge, and improving the film quality will be necessary to harness the full potential of our materials. Hence, we further investigated energy storage application using the bulk COF materials.

Energy Storage Properties. The interplay between aromatic and antiaromatic states can enable a defined redox activity in carbon-rich materials, which is usually only observed for compounds containing heteroatoms and localized redox-active groups. For DBP derivatives, this is manifested in ambipolar electrochemical behavior and both accessible oxidation and reduction states,[26,35] which, however, have not been explored in charge storage applications. Few other examples exist, such as an antiaromatic norcorrole used as battery electrode material[8] or a purely hydrocarbon para-cyclophane-1,9,17,25-tetraene,[86] switching from local to global aromaticity upon reduction. To evaluate the energy storage capabilities of the antiaromatic DBP-COFs reported herein, their electrochemical properties were investigated in Li-organic half cells. Composite electrodes consisting of 30 wt % COF or POP, 50 wt % acetylene black as conductive carbon additive, and 20 wt % PVdF binder on Al or Cu foil were investigated in coin cells vs. Li as the counter electrode, using 1 M $LiPF_6$ in EC/DMC 1:1 as the electrolyte. Of all DBP materials investigated, the 2,7-linked **PhDBP-TFP** COF and the 5,10-connected **DBP-Ph-TFP** POP yielded the best results, which will be discussed in the following. The materials were investigated as negative or positive electrode materials. The results of using **PhDBP-TFP** COF and **DBP-Ph-TFP** POP as negative electrode materials in Li-organic batteries in the potential range of 3.0−0.05 V vs. Li/Li^+ (Figure S76) as well as electrochemical impedance spectra (Figures S81 and S82) are shown in the Supporting Information.

As a positive electrode material in a dual-ion Li-organic battery, the cyclic voltammograms (CVs) of **PhDBP-TFP** COF vs. Li/Li^+ showed a redox event above 3.8 V (Figure 7a). This redox potential of the COF corresponds well to the oxidation of the DBP small molecules **3** and **5** as COF precursors, which confirms that the redox activity stems from the DBP units (Figures S65 and S67). In the CV, **PhDBP-TFP** COF shows a higher integral charge area compared to **DBP-Ph-TFP** POP, also reflected in the galvanostatic cycling experiments. The specific discharge capacity of **PhDBP-TFP** COF (26 mA h g^{-1}) at 100 mA g^{-1} is much higher than that of **DBP-Ph-TFP** POP with 14 mA h g^{-1} (Figure 7b). These values correspond to 47 and 25% of the theoretical specific capacities (54.8 mA h g^{-1}) of COF and POP, respectively, assuming a one-electron oxidation. In both cases, a plateau centered around 3.95 V vs. Li/Li^+ is visible, a high value for COFs.[64,87,88] We attribute the electrochemical performance to the active COF/POP material as the pure acetylene black as conductive carbon contributes only 2 mA h g^{-1} to the specific capacity in the same potential range (Figure S78). Both **PhDBP-TFP** COF and **DBP-Ph-TFP** POP show good rate performance, with only small capacity losses when increasing the C-rate from 0.4 to 10 C, and even rates as high as 20 and 40 C resulted in noticeable specific capacities (Figure 7c). Decreasing the C-rates to 0.4 C restored the initial specific capacities.

To investigate the role of the antiaromatic DBP units on the battery performance of COF and POP and to shed light on the differences in their performance, we performed DFT calculations on subunits of **PhDBP-TFP** COF and **DBP-Ph-TFP** POP, denoted as **PhDBP-TFP*** and **DBP-Ph-TFP***, respectively (Supporting Information, Section S18 for exact structures; PW6B95-D3/def2-QZVP(COSMO)//B3LYP-D3/def2-TZVP(COSMO) level of theory). The calculations considered a redox equation for the reversible oxidation of a DBP to its radical cation (Figure 7d). In **PhDBP-TFP***, the oxidation site is revealed by the localized HOMO on the DBP core (Figure 7e). In **DBP-Ph-TFP***, on the other hand, the HOMO is largely delocalized from the DBP over the 5,10-biaryl substituents to the COF node, and hence, the redox activity is not centered solely on the antiaromatic DBP core. This is also reflected in the (de)localization of the spin density in the radical cations of **PhDBP-TFP*$^{•+}$** and **DBP-Ph-TFP*$^{•+}$** (Figure 7f) as well as the electrostatic potential map of these oxidized species (Figure 7g). In the **PhDBP-TFP*$^{•+}$** radical cation, the spin density and the charges are evenly distributed over the DBP core, showing that the redox activity is centered on the DBP core. In the **DBP-Ph-TFP*$^{•+}$** radical cation, on the other hand, the spin density is distributed only on the central part of the DBP as well as the 5,10-connected aryl linkages. The electrostatic potential maps are in agreement, indicating the minor role the DBP core plays in the redox activity of this material.

An aromaticity analysis based on $NICS_{zz}(1.7)$[89] values of the neutral and radical cation form of **PhDBP-TFP*** and **DBP-Ph-TFP*** gave further insights into the stabilization in the charged state of the material (Figure 7a). Upon oxidation of **PhDBP-TFP***, the antiaromatic character in the 5-membered DBP rings decreases with a change in $NICS_{ZZ}(1.7)$ from +2.9 to −1.2 ppm, while the aromaticity in the 6-membered rings remains constant ($NICS_{zz}$ −5.2 to −5.4 ppm). This confirms

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J. Am. Chem. Soc. 2023, 145, 2840−2851

the initially proposed stabilization of the oxidized DBP. In **DBP-Ph-TFP***, however, the antiaromatic character in the 5-membered DBP rings increases upon oxidation ($NICS_{zz}(1.7)$ changes from 4.5 to 6.8 ppm) concurrently with a decrease in aromaticity of the 6-membered rings ($NICS_{zz}(1.7)$ −11.0 to −9.8 ppm). These calculations show that the antiaromatic DBP core in the **PhDBP-TFP** COF plays a major role in the redox activity of the COF, while in **DBP-Ph-TFP** POP, the redox activity is not strongly related to the DBP core. Since the materials are otherwise structurally similar, we attribute the better battery performance of the **PhDBP-TFP** COF to this effect of antiaromaticity, which is less present in the **DBP-Ph-TFP** POP, resulting in a reduced battery performance.

To shed further insights into the role of antiaromaticity on the battery performance, we investigated the aromatic **BND-TFP** COF (in some reports referred to as **TpBd** COF[90]) for comparison. **BND-TFP** COF is structurally analogous to **PhDBP-TFP** COF, but the DBP units are replaced by biphenyl groups, which lack antiaromaticity (structures in the Supporting Information, Figure S79). **BND-TFP** COF[81] was synthesized according to the literature. The **BND-TFP** COF electrodes gave CV profiles showing a moderate redox activity between 3.6 and 4.2 V vs. Li/Li^+ (Figure S80). However, the aromatic **BND-TFP** COF provided specific discharge capacities of only 19 mA h g^{-1} in the second galvanostatic cycle, lower than those obtained for the antiaromatic **PhDBP-TFP** COF of 26 mA h g^{-1} (see above). Hence, the antiaromatic DBP units in the **PhDBP-TFP** COF significantly increased its battery performance compared to the aromatic **BND-TFP** COF, further confirming the positive effect antiaromaticity can have in electrode-active COFs.

In summary, the antiaromatic **PhDBP-TFP** COF and **DBP-Ph-TFP** POP both showed electrochemical activity as positive electrode materials in Li-organic batteries. However, the performance of **PhDBP-TFP** COF was significantly better due to the localized redox activity at the antiaromatic DBP core, as shown by DFT calculations and experimental comparison with the structurally similar but aromatic **BND-TFP** COF.

■ CONCLUSIONS

While aromatic building blocks are well established in framework materials, antiaromatic units have been widely neglected. We report the first framework materials constructed from antiaromatic building blocks. We have developed three highly porous and crystalline COFs and two porous organic polymers (POPs) based on dibenzopentalene. The antiaromatic materials show broad optical absorption of up to 800 nm. **PhDBP-TFP** COF is semiconductive (4×10^{-8} S cm^{-1}) in the doped state and shows distinct photoconductivity in the range of 375−660 nm LED irradiation. Owing to their antiaromatic character, DBPs are stabilized in their oxidized and reduced states, making them potential candidates for battery applications. Using crystalline **PhDBP-TFP** COF as a positive electrode material in Li-organic batteries, a larger capacity of 26 mA h g^{-1} at a high potential of 3.9 V vs. Li/Li^+ was found compared to amorphous **DBP-Ph-TFP** POP with 14 mA h g^{-1}. DFT calculations and experimental comparison to the structurally similar but aromatic **BND-TFP** COF showed that this enhanced performance of **PhDBP-TFP** COF can be attributed to the localized redox activity at the antiaromatic DBP core, even compared to **DBP-Ph-TFP** POP, in which the DBP units are incorporated in a different fashion. Overall, this work demonstrates that antiaromaticity offers a new design principle for framework materials in energy storage applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.2c10501.

Detailed information on experimental methods, materials used, synthetic characterization, calculations, and data (PDF)

Accession Codes

CCDC 2082679, 2098056, and 2129185 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.
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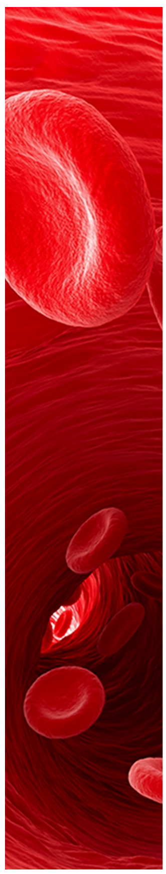

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